Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Software Engineer with **7+ years in multifamily PropTech** (Entrata -- affordable housing & residential SaaS, same industry as ApartmentIQ) and 9+ years total experience spanning data pipelines, full-stack development, and AI-powered applications. Built production ETL systems processing 1.6M+ HUD records, async pipelines (RabbitMQ), REST API integrations, and React/Python/PostgreSQL features across a multi-tenant platform with 100K+ monthly active users. At Intuit, integrating LLMs (Claude, GPT) into financial workflows using LangGraph, RAG, and agentic orchestration. Built LeaseI -- an 8-agent AI system for **lease compliance** -- and AgentProbe for multi-provider LLM routing. Strong Python, FastAPI, React/TypeScript, PostgreSQL, AWS (Lambda, S3, RDS, SQS, DynamoDB, Kinesis), Docker, Kubernetes, and Terraform. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / FastAPI / SQL | React / TypeScript / JavaScript | ETL Pipelines / Data Ingestion | REST APIs / Backend Systems | LLM Integration (Claude, GPT, Llama) | Agentic Workflows (LangGraph) | RAG Pipelines / Vector Retrieval | Human-in-the-Loop Escalation | ML Pipelines (scikit-learn, XGBoost) | AWS (Lambda, S3, RDS, SQS, DynamoDB, Kinesis) | Terraform (IaC) | Docker / Kubernetes | PostgreSQL / Advanced SQL | CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry) | Multifamily PropTech Domain

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- AI/LLM Tools & Production Systems -- Virginia, USA

- Build and deploy **LLM-based tools** (Claude, GPT) for financial data workflows using the Anthropic Messages API and Agent SDK -- prompt engineering, tool use, structured output, and guardrails on AWS.
- Develop **full-stack features** with Python/FastAPI backends and React/TypeScript frontends; own POC-to-production lifecycle with golden-dataset eval harnesses and Prometheus/OpenTelemetry instrumentation.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through scoring APIs; deploy via Docker and GitHub Actions CI/CD.
- Implement **agentic orchestration** (LangGraph) with human-in-the-loop escalation patterns for regulated financial data; daily practitioner of AI-assisted development with rigorous review before shipping.

Entrata -- Multifamily & Affordable Housing SaaS

2017 -- Jul 2024

Senior Software Engineer (Oct 2020 -- Jul 2024) | Junior Software Engineer (2017 -- Oct 2020) -- Multi-tenant platform, 100K+ monthly active users

- Delivered 20+ features end-to-end across **multifamily property management, rent calculation, and affordable housing compliance** workflows (Python, PHP, Java, React, PostgreSQL, Oracle) -- directly parallel to ApartmentIQ's product domain.
- Maintained the annual **HUD income-limit ETL pipeline**: ingested federal XML/JSON data from external APIs, parsed and reconciled ~180,000 records per cycle with validation and rollback, deployed on AWS (Lambda, S3, RDS, SQS).
- Built **async data processing** (RabbitMQ) for rent updates across thousands of tenant records with idempotency guarantees and dead-letter handling; optimized 200+ SQL queries via execution plan analysis (30% performance gain).
- Designed **Docker microservices architecture** (50% better scalability, 30% faster deployments); deployed on AWS with Terraform IaC; instrumented services with Prometheus and OpenTelemetry for production observability.
- Built large-scale **data import and reconciliation workflows** (1.6M+ HUD / Tax Credit records); led AWS S3 migration from legacy storage with Libsodium encryption for resident PHI; created 56 state-specific housing certification documents.
- Reviewed 300+ PRs; mentored junior engineers on distributed systems and data pipeline patterns; built Python/Bash automation eliminating 80% of manual operational workload.

PROJECTS

Leasel -- AI Lease Compliance Platform (8-Agent RAG System)

Python · LangGraph · Agent SDK · FastAPI · RAG · PostgreSQL

8-agent LangGraph orchestration system for **apartment lease compliance** -- directly relevant to ApartmentIQ's domain. Full POC-to-production pattern: triage, hybrid-search RAG retrieval (TF-IDF + semantic), rule evaluation, confidence scoring, and human escalation at defined thresholds. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- AI Evaluation & Multi-Provider Routing Infrastructure

Python · FastAPI · Provider Routing · Cost Management · CI/CD

15-endpoint FastAPI service with provider routing across five LLM/ML inference backends (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Scoring engine evaluates output quality on five metrics with cost-aware routing; CI regression against golden datasets. Addresses AI platform observability, cost management, and flexible model adoption.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical NLP API

Python · Flask · React · TypeScript · REST APIs · RBAC · Docker

RESTful API translating clinical English/Spanish queries into FHIR data calls using LLM tools and BERT. Fine-grained RBAC, containerized deployment, and automated testing -- regulated-data AI integration pattern with React/TypeScript frontend.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 I GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017